

Worksheet 4: Arithmetic and Alternating Sequences

Topic Focus: Constant addition/subtraction, mixed rules, and finding missing terms.

1. Find the missing number: 12, 19, 26, 33, __, 47.

A) 39

B) 40

C) 41

D) 42

Hint: Find the common difference by subtracting the first term from the second ($19 - 12 = 7$).

2. What is the rule for this sequence? 100, 95, 89, 82, 74, ...

A) Subtract 5 each time

B) Subtract an increasing number (5, 6, 7...)

C) Subtract a decreasing number

D) Divide by 2

Hint: Analyze differences: -5, -6, -7, -8. The gap gets wider.

3. Look at this alternating sequence: 1, 10, 2, 9, 3, 8, __, __. Next two numbers?

A) 4, 7

B) 4, 9

C) 5, 6

D) 4, 10

Hint: Treat it as two sequences mixed together. Seq 1: 1, 2, 3... Seq 2: 10, 9, 8...

4. Which number does NOT belong in this sequence of multiples? 7, 14, 21, 28, 36, 42.

A) 21

B) 28

C) 36

D) 42

Hint: Check divisibility. All numbers should be perfectly divisible by 7.

5. Find the rule and next number: 2, 6, 12, 20, 30, __.

A) 40 (Add 10)

B) 42 (Add even numbers 4, 6, 8, 10, 12)

C) 36 (Add 6)

D) 48 (Double it)

Hint: Look at the gaps: +4, +6, +8, +10. The next gap must be +12.

6. What is the 10th even number?

A) 10

B) 18

C) 20

D) 22

Hint/concept: The rule for the 'nth' even number is simply $2 \times n$.

7. Look at the sequence: 5, 15, 45, 135, ... What is the rule?

A) Add 10

B) Multiply by 3

C) Add 30

D) Square it

Hint/thinking process: When gaps grow very large very quickly, test for multiplication (Geometric progression).

8. Find the missing term: 2, 3, 5, 7, 11, __, 17, 19.

A) 12

B) 13

C) 14

D) 15

Hint: These numbers only have two factors: 1 and themselves. This is the prime number sequence.

9. A train arrives every 15 mins. First train is at 9:05 AM. When does the fourth train arrive?

A) 9:45 AM

B) 9:50 AM

C) 10:00 AM

D) 10:05 AM

Hint: Add 15 mins three times (for the 2nd, 3rd, and 4th trains). $9:05 + 45 \text{ mins}$.

10. In sequence 1, 4, 2, 8, 4, 16, 8, ... what two rules alternate?

A) Add 3, subtract 2

B) Multiply 4, divide 2

C) Multiply 2, add 4

D) Square it, halve it

Hint: Check the operations between consecutive numbers. $1 \times 4 = 4$. $4 / 2 = 2$. $2 \times 4 = 8$. $8 / 2 = 4$.

11. Sequence of fractions: $1/2$, $2/3$, $3/4$, $4/5$, ... What is the 10th fraction?

A) $9/10$

B) $10/11$

C) $11/12$

D) $10/12$

Hint: The numerator is the position number. The denominator is the position number + 1.

12. What comes next? 10, 20, 11, 22, 12, 24, 13, __

- A) 25 B) 26 C) 14 D) 28

Hint: Two alternating sequences. 10, 11, 12, 13... and 20, 22, 24... Next is the even sequence.

13. A sequence goes: 50, 43, 36, 29... What is the first negative number in this sequence?

- A) -1 B) -3 C) -6 D) -7

Hint: Keep subtracting 7. 29 -> 22 -> 15 -> 8 -> 1 -> -6.

14. What is a quick way to find the sum of $1+2+3+\dots+10$?

- A) Multiply 10×10 B) $(10 \times 11) / 2$ C) Add only the evens D) It's impossible to do quickly

Hint: The formula for the sum of the first N natural numbers is $n(n+1)/2$.

15. Identify the pattern: 3, -6, 12, -24, 48, __

- A) -96 B) 96 C) -72 D) 72

Hint: The numbers double in magnitude, but flip signs. Multiply by -2 each step.